Epidemic model on a network: analysis and applications to COVID-19

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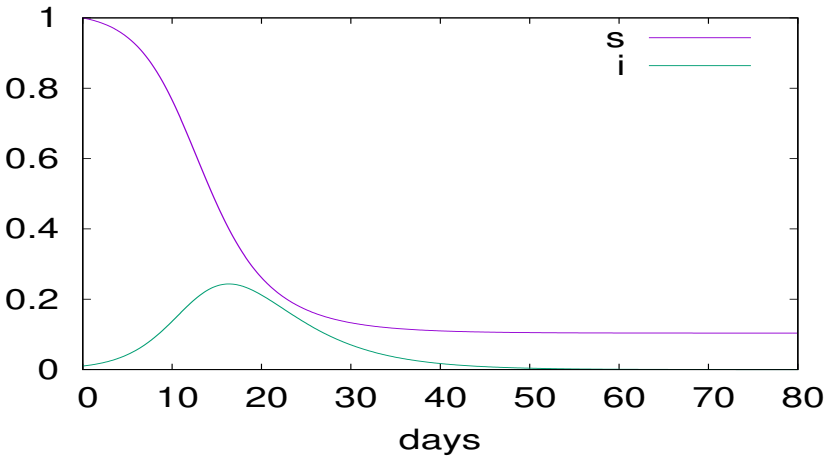
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Outline

- COVID 19 : the data
- The model: SIR + diffusion on graph
- Isolation strategy
- Arrival time of COVID to Mexico
- Deconfinement strategy

Kermack-McKendrick SIR model

$$\begin{cases} \dot{S} = -\alpha SI, \\ \dot{I} = \alpha SI - \beta I \\ \dot{R} = \beta I \end{cases} \tag{1}$$



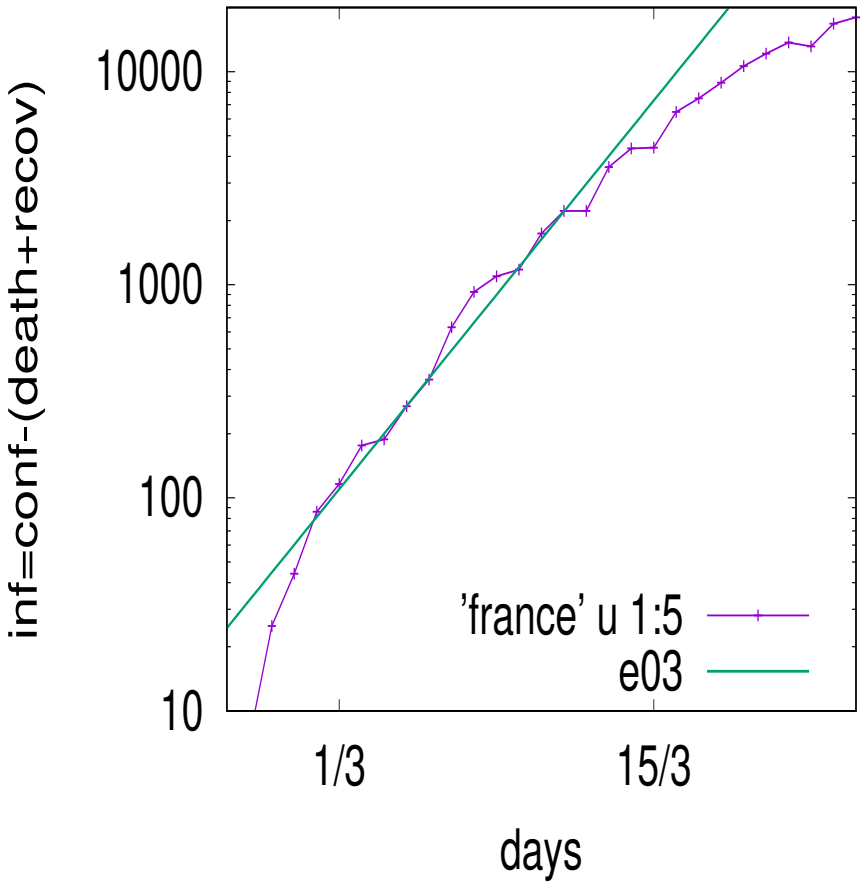
$R_0 \equiv \frac{\alpha}{\beta}$

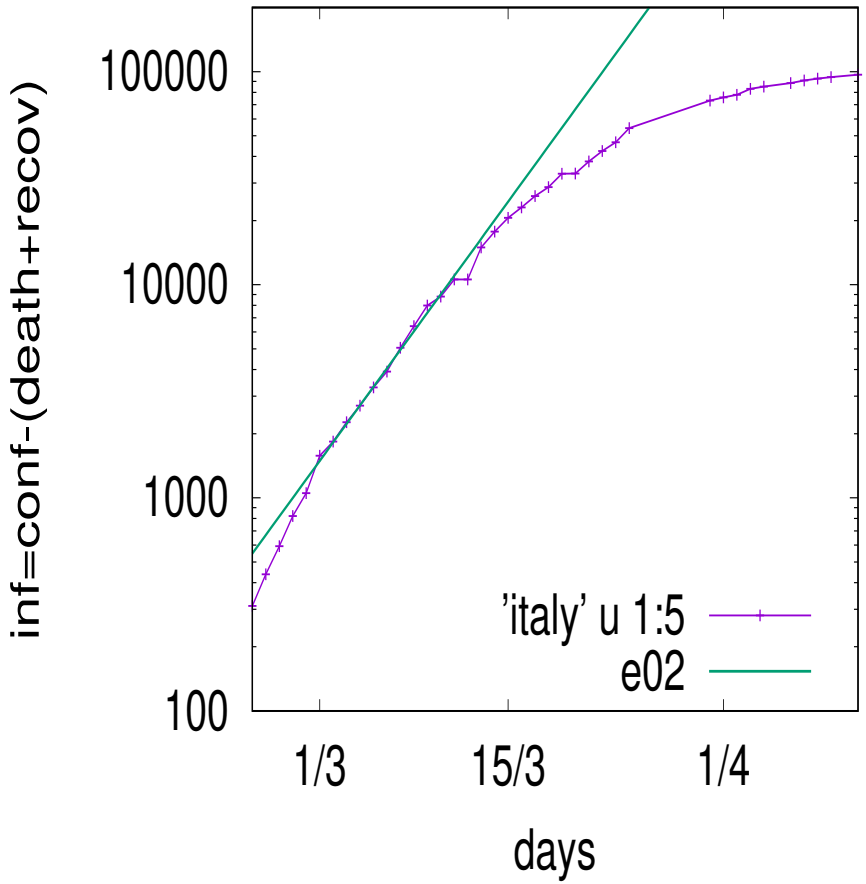
Peak of infection (S^*, I^*)

$S^* = \frac{1}{R_0}, \quad I^* = I_0 + S_0 - \frac{1}{R_0}(1 + \log(R_0 S_0))$

Data COVID 19 : expected behavior

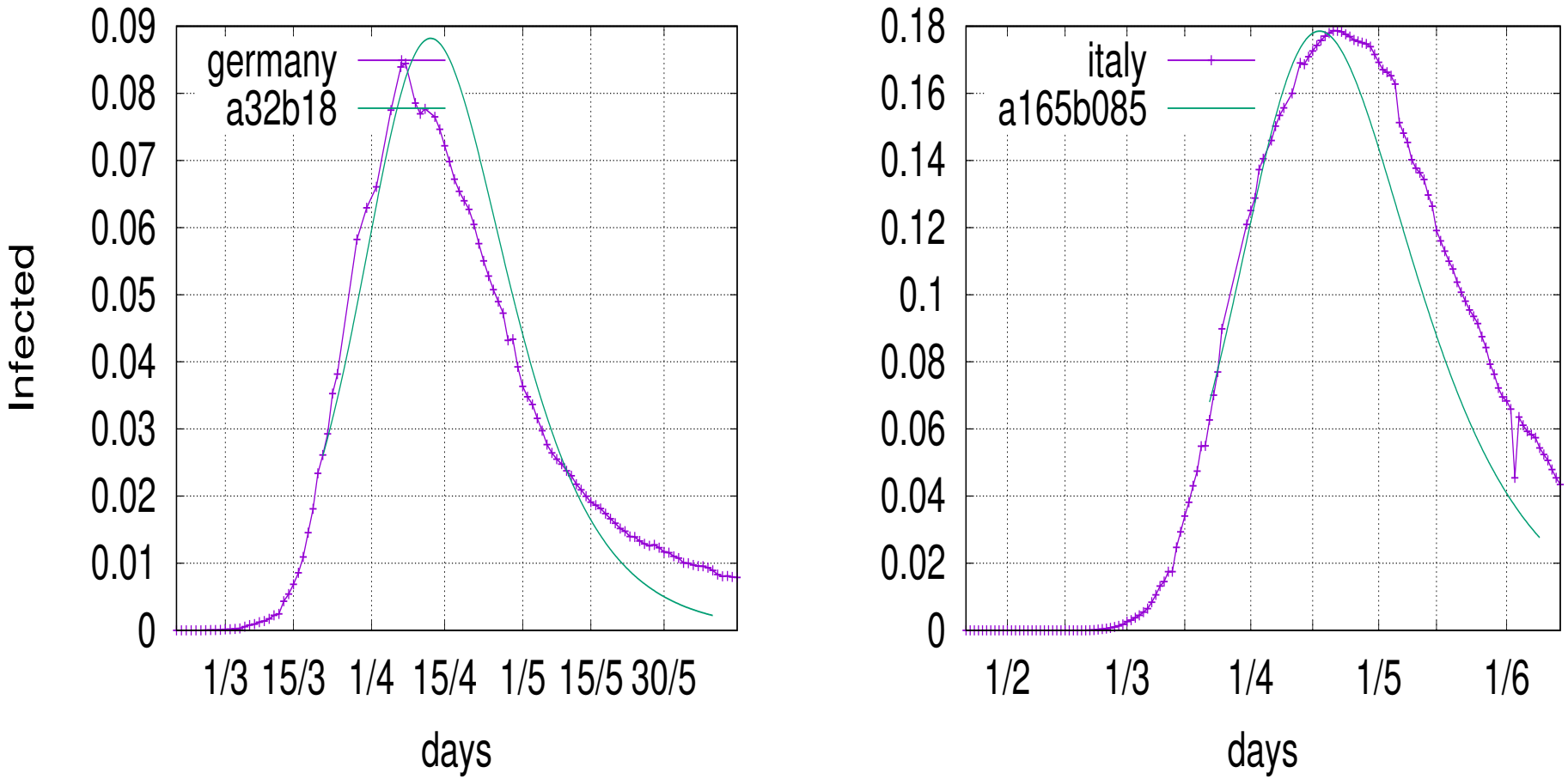
Data : J. Hopkins and Worldometer



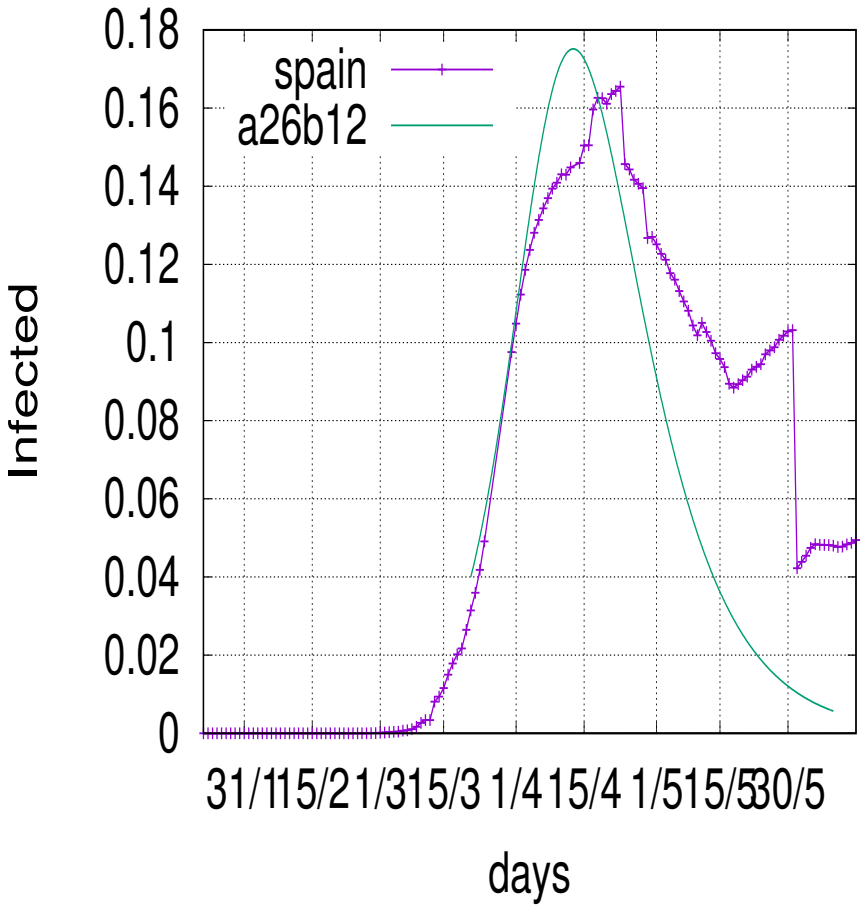


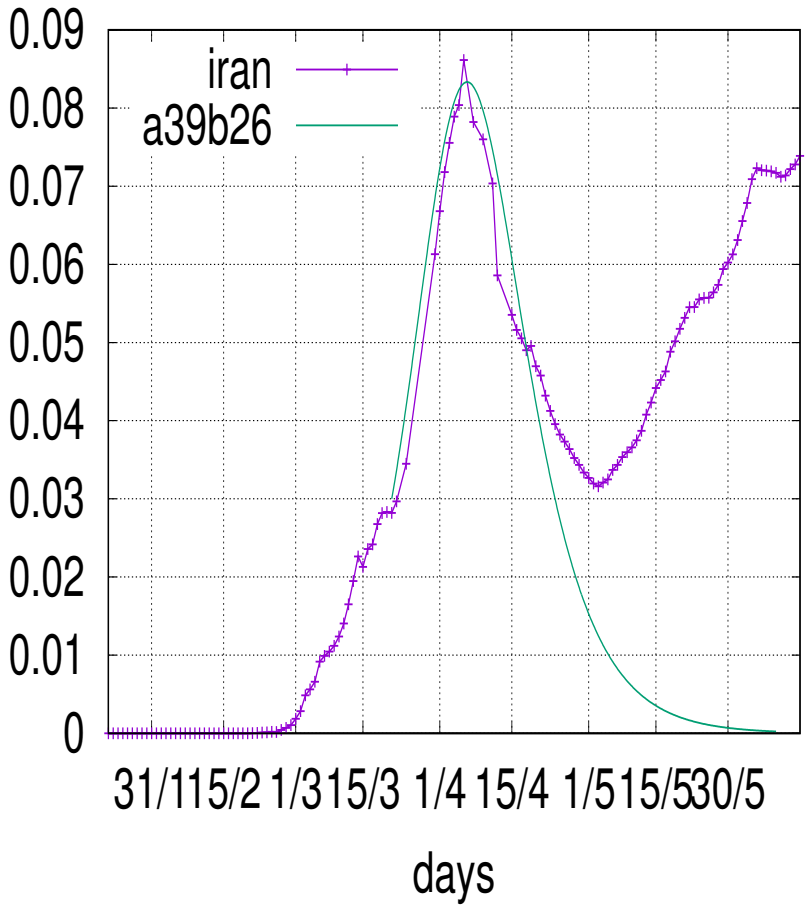
Data COVID 19 : expected behavior

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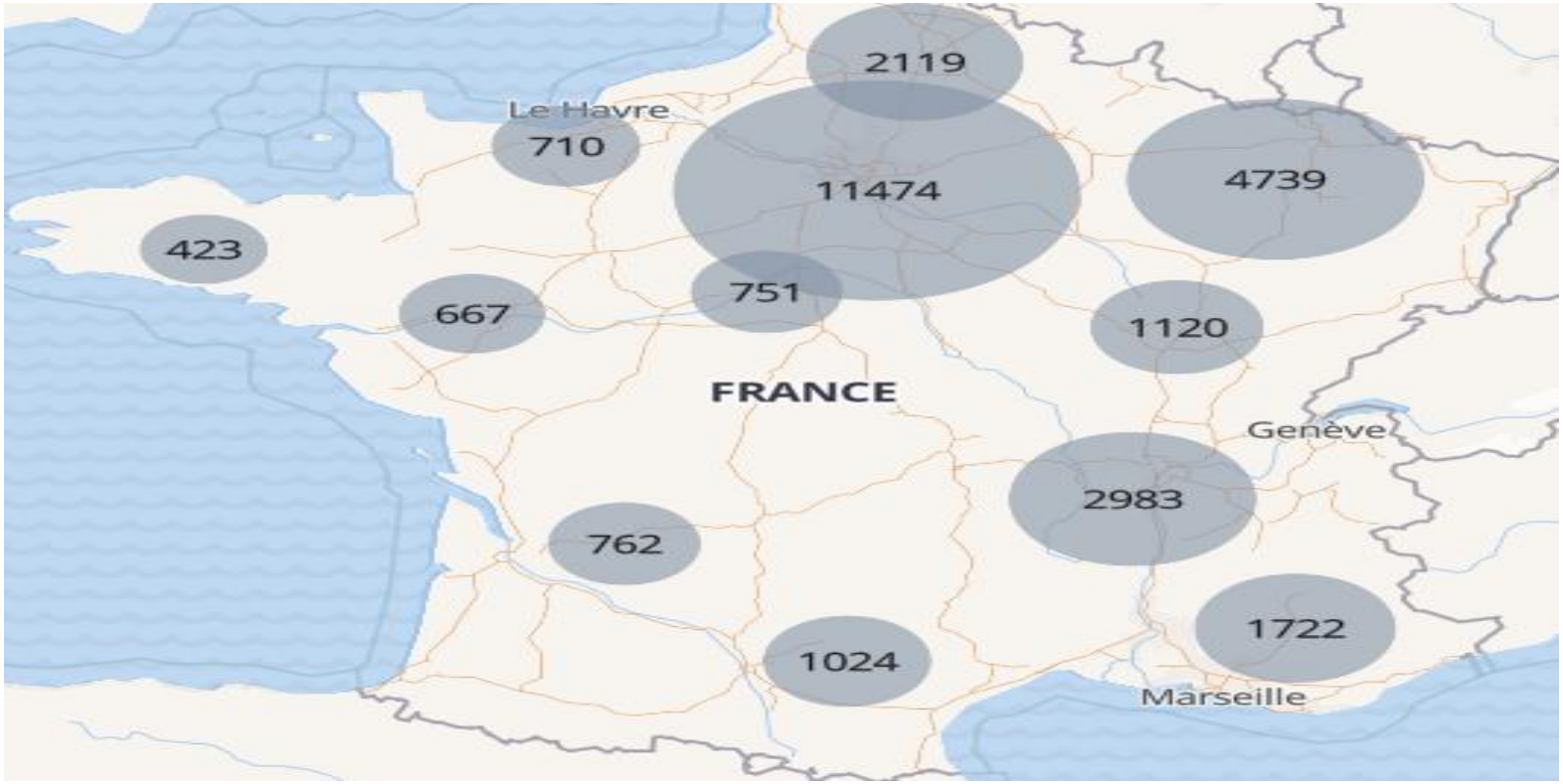


Data COVID 19 : unexpected behavior





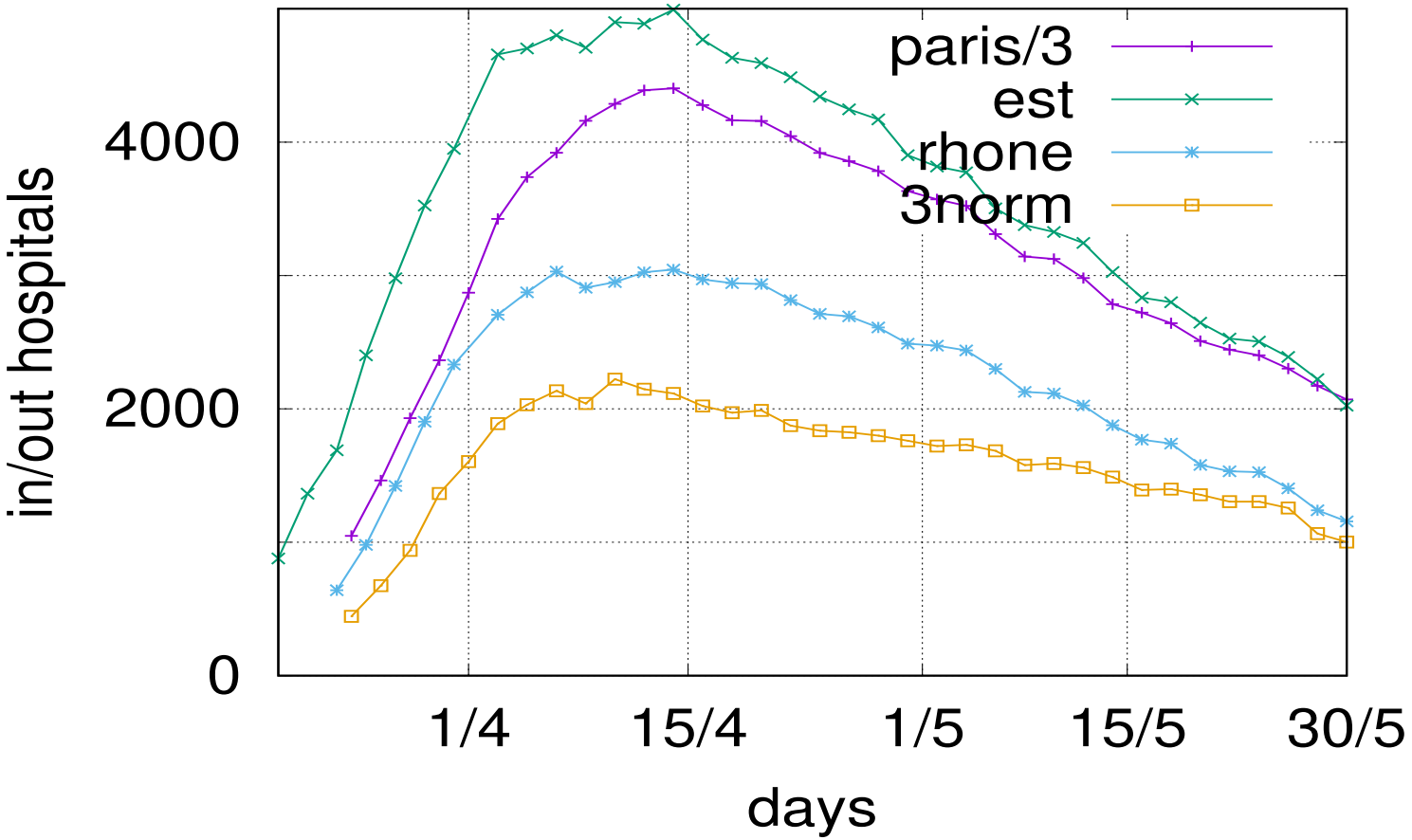
Possible explanation : regional data FRANCE COVID 19



Hospital admissions April 5 2020

https://veille-coronavirus.fr/

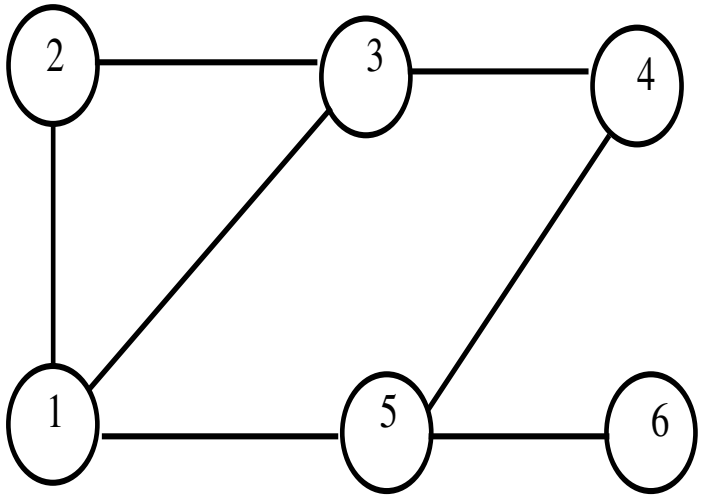
Possible explanation : regional data FRANCE COVID 19



Hospital in/out https://veille-coronavirus.fr/

The graph model: SIR + geographic diffusion



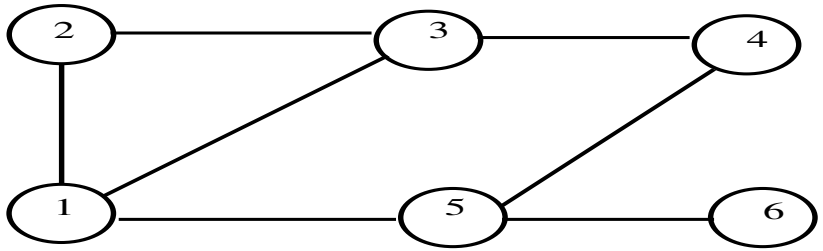


$$\dot{S}_j = -\alpha \frac{S_j}{N_j} I_j + \epsilon \sum_{k \sim j} (S_k - S_j), \tag{2}$$

$$\dot{I}_j = \alpha \frac{S_j}{N_j} I_j - \beta I_j + \epsilon \sum_{k \sim j} (I_k - I_j), \tag{3}$$

Well posed and positive , Brockmann & Helbling 2013

Graph Laplacian

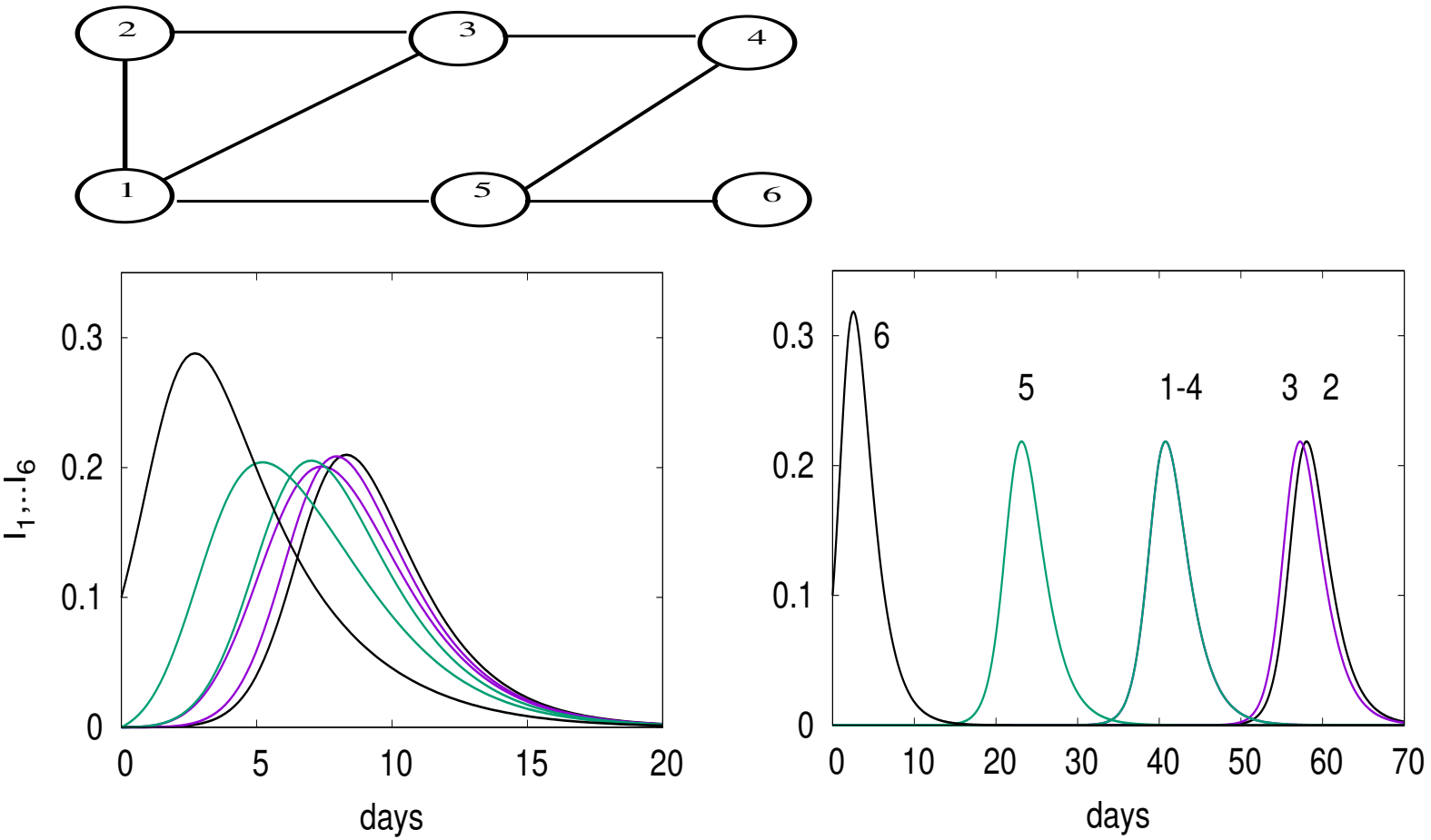


$$\begin{cases} \dot{S} = \epsilon \Delta S - \alpha S\ I, \\ \dot{I} = \epsilon \Delta I + \alpha SI - \beta I. \end{cases}$$

$S = (S_1, S_2, \ldots, S_n)^T, \quad I = (I_1, I_2, \ldots, I_n)^T,$
$\alpha \equiv (\alpha/N_1, \alpha/N_2, \ldots, \alpha/N_n)^T,$
$SI \equiv (S_1 I_1, S_2 I_2, \ldots, S_n I_n)^T$. Δ graph Laplacian matrix

$$\Delta = \begin{bmatrix} -3 & 1 & 1 & 0 & 1 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 1 & 0 & 0 & 1 & -3 & 1 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix}.$$

Influence of diffusion



Time evolution $I_k(t),\ k = 1, \ldots, 6$ for an outbreak at node 6 for $\epsilon = 0.1$ (left panel) and 10^{-7} (right panel).
$\alpha = 1.5$ and $\beta = 0.625$.

Epidemic criterion

$$\dot{I} = AI$$

$$A = \epsilon\Delta - \beta Id_n + \mathrm{diag}(\alpha_1 S_1, \alpha_2 S_2, \ldots, \alpha_n S_n)$$

Epidemic criterion : onset of epidemic if one eigenvalue σ_i of A is positive.

$\epsilon << 1$ each node develops independently

Vaccination or isolation of one node

$S = 0.26 \times (1,1,1,1,1,1)^T$ and $S_j = 0$ at a specific node j ,

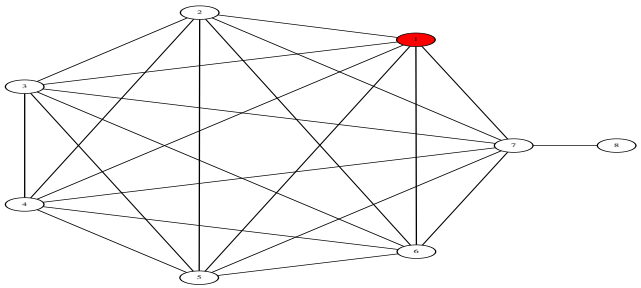
j	degree	σ_3	σ_2	σ_1
6	1	$-2.46\ 10^{-1}$	$-1.30\ 10^{-1}$	$1.22\ 10^{-2}$
2	2	$-2.14\ 10^{-1}$	$-7.85\ 10^{-2}$	$3.83\ 10^{-3}$
4	2	$-2.51\ 10^{-1}$	$-4.79\ 10^{-2}$	$-1.06\ 10^{-3}$
1	3	$-1.52\ 10^{-1}$	$-6.41\ 10^{-2}$	$-6.72\ 10^{-3}$
3	3	$-1.44\ 10^{-1}$	$-8.29\ 10^{-2}$	$-3.71\ 10^{-3}$
5	3	$-1.52\ 10^{-1}$	$-6.41\ 10^{-2}$	$-6.72\ 10^{-3}$

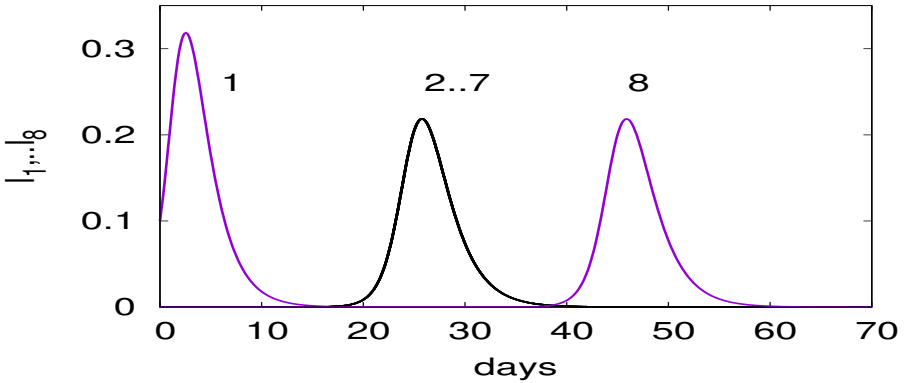
Table: Vaccinated node j and associated eigenvalues of A

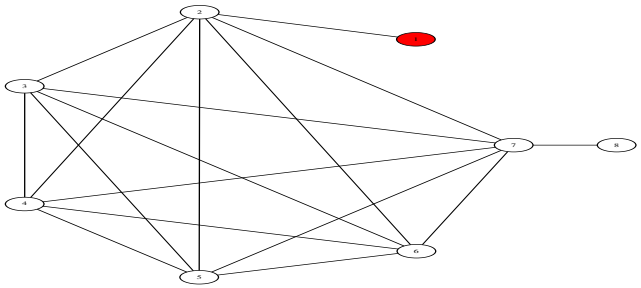
In the absence of vaccination, the largest eigenvalue of the matrix A is $2.5\ 10^{-2}$.

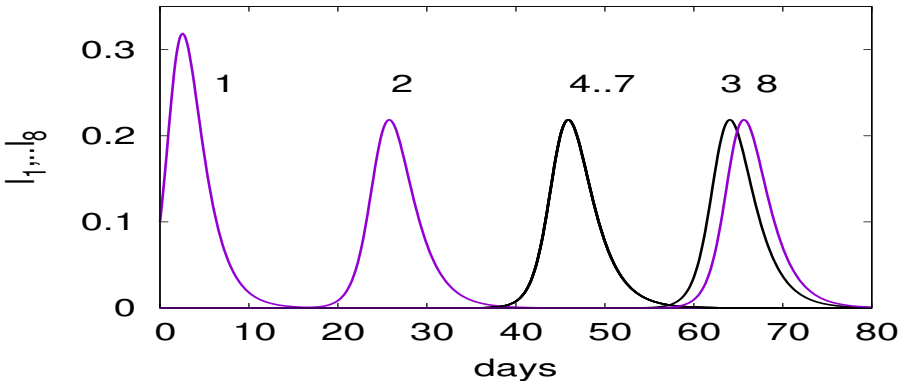
Most effective to vaccinate nodes 1,3 and 5 (highest degree)

Start of epidemic in Mexico : $\approx$ March 20









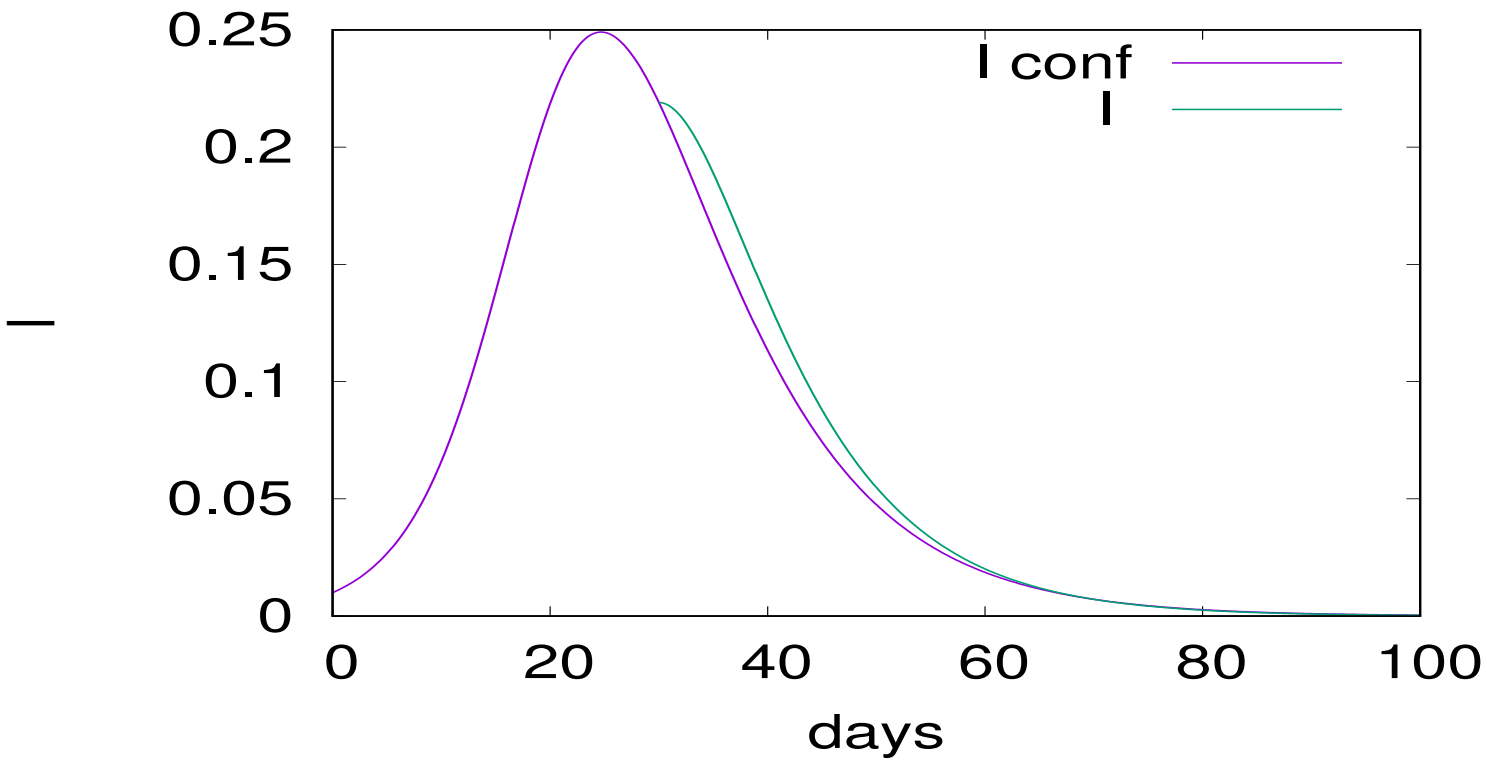
7 nodes forming a complete graph Hubei, Beijing, Shanghai, Japan, western Europe, eastern USA and western USA
8th node is Mexico

$\alpha = 0.5, \beta = 0.2$ and $\epsilon = 10^{-6}\text{days}^{-2}$

Confinement

(i) reduce capacity N_j of each node j and therefore α_j
(ii) reduce diffusion ϵ, ie the travel between nodes

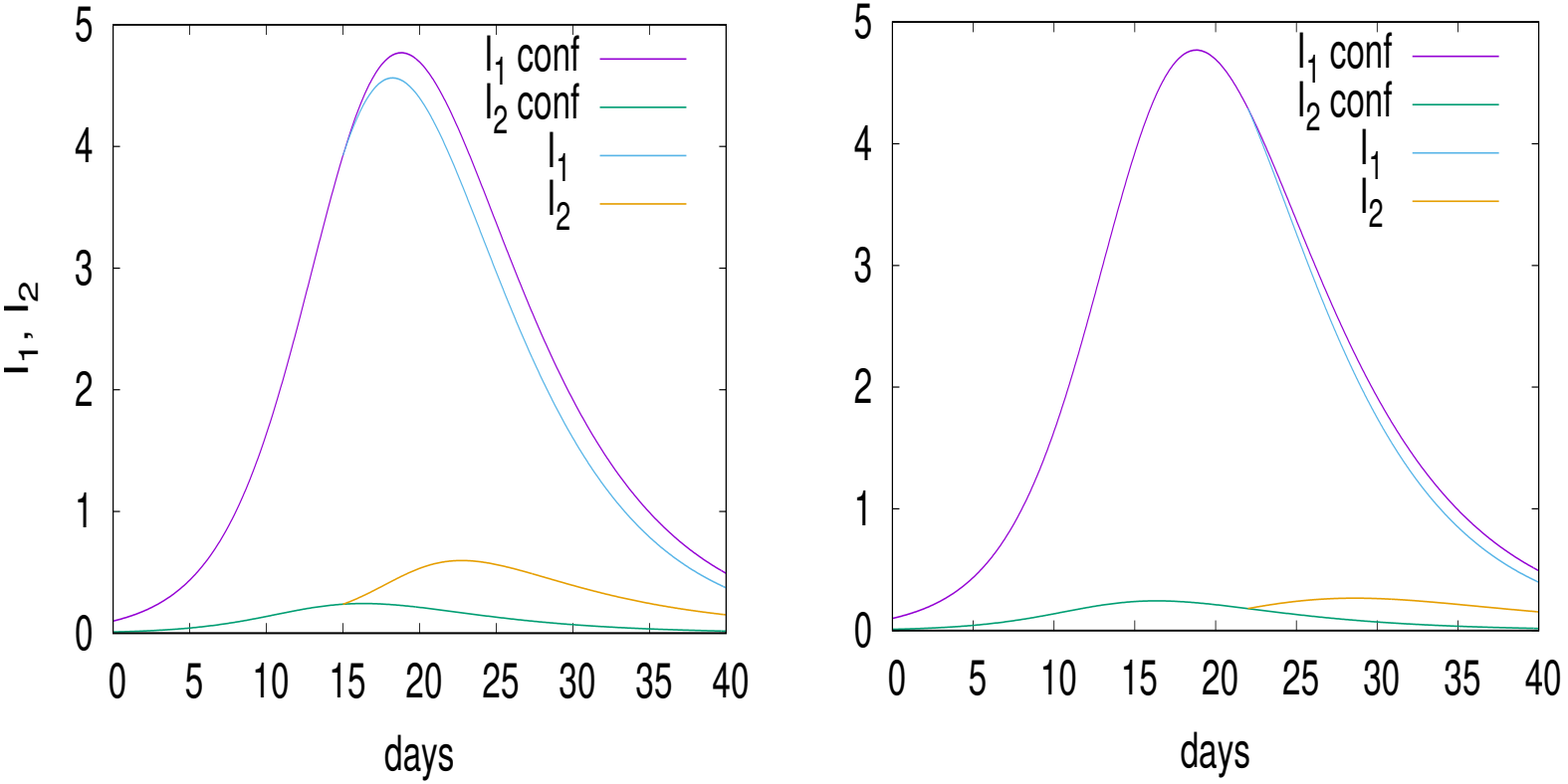
Deconfinement strategy : increase α



Deconfinement by increasing α at a single node.

$\alpha = 0.33, \quad t < 30, \quad \alpha = 0.5, \quad t > 30,$

Deconfinement strategy : increase ϵ



Deconfinement by increasing ϵ in a two node graph
increase of diffusion at $t = 15$ (left) and $t = 22$ (right)

Conclusions and recommendations

- Analyzed a model of an epidemic on a network,
  essential ingredients : importation of infected subjects,
  minimum number of parameters
- Generalized **epidemic criterion** of Kermack-McKendrick
  small diffusion, outbreaks occur at different times
  large diffusion, outbreaks synchronize
  Best to **isolate high degree nodes**
- **Recommendations** : confine early, masks
  Control borders , avoid super-spreading events
  Deconfine after peak and slowly

F. Bustamante-Castañeda et al, https://arxiv.org/abs/1906.07449
G. Cruz-Pacheco et al, Clinical and translational investigation, 2020